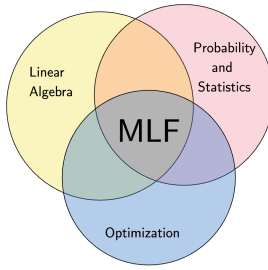


Week-4, Session-2



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Disclaimer: These notes were generated as a part of this [live session](#). It can't be used as a standalone resource or a set of notes for this week.

1. Recap

- Eigenvalues and eigenvectors

$$Ax = \lambda x$$

- Characteristic polynomial

$$|A - \lambda I| = 0$$

- Computing eigenvectors

$$(A - \lambda I)v = 0$$

2. Diagonal matrices

A diagonal matrix is given by $D = \text{diag}(a_1, \dots, a_n)$.

$$D = \begin{bmatrix} a_1 & & \\ & \ddots & \\ & & a_n \end{bmatrix}$$

Determinant

$$|D| = a_1 \cdots a_n$$

Matrix power

$$D^k = \text{diag}(a_1^k, \cdots, a_n^k)$$

Invertibility and Rank

A diagonal matrix is invertible if:

- $|D| \neq 0$
- None of the diagonal entries are zero
- All diagonal entries are non-zero

If D is invertible, then:

$$D^{-1} = \text{diag}\left(\frac{1}{a_1}, \cdots, \frac{1}{a_n}\right)$$

The rank of a diagonal matrix is the number of non-zero diagonal entries.

Eigenvalues

The characteristic polynomial is just:

$$|D - \lambda I| = (a_1 - \lambda) \cdots (a_n - \lambda)$$

Therefore, the eigenvalues of a diagonal matrix are the entries on the main diagonal.

Eigenvectors

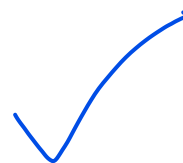
If a_i is the i^{th} diagonal entry of D , then e_i is an eigenvector of D corresponding to eigenvalue a_i , where $e_i = (0, \cdots, 1, \cdots, 0)$ is the i^{th} element in the standard basis. For example:

$$De_2 = \begin{bmatrix} a_1 & & \\ & a_2 & \\ & & a_3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ a_2 \\ 0 \end{bmatrix}$$

$$= a_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

3. Diagonalizability: Definition-1



Diagonal matrices have very useful properties. Can we find matrices that are **similar** to diagonal matrices? Can we find matrices that share some of these properties of diagonal matrices, at least indirectly?

Definition-1: A matrix A of shape $n \times n$ is called **diagonalizable** if it is similar to a diagonal matrix of the same shape.

Remark-1: Two matrices A and B of shape $n \times n$ are similar if there is some invertible matrix P of the same shape such that:

$$B = PAP^{-1}$$

Remark-2: Similar matrices actually represent the same linear transformation in different bases. One can view the invertible matrix P as a change-of-basis matrix. This is what is called a similarity transformation. For example, consider:

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(x, y) = (x + y, x - y)$$

Basis: $\beta = \{(1, 0), (0, 1)\}$

$$A = [T]_{\beta}^{\beta} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Basis: $\gamma = \{(1, 1), (-1, 1)\}$

We now need to express $T(1, 1) = (2, 0)$ and $T(-1, 1) = (0, -2)$ in the new basis γ . We see that $(2, 0) = (1, 1) - (-1, 1)$ and $(0, -2) = -(1, 1) - (-1, 1)$. Therefore:

$$B = [T]_{\gamma}^{\gamma} = \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$$

A and B are similar. Meaning, there is some invertible matrix that connects A and B . The columns of this invertible matrix is actually the elements of the basis γ :

$$P = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

Now, we can verify that:

$$A = PBP^{-1}$$

Diagonalizable matrices:

$$A = PDP^{-1}$$

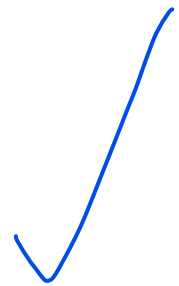
We will study the consequences of this definition. If we right-multiply by P on both sides:

$$AP = PD$$

$$P = \begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix}, \quad D = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{bmatrix}$$

$$AP = \begin{bmatrix} | & | & | \\ Av_1 & Av_2 & Av_3 \\ | & | & | \end{bmatrix}$$

$$PD = \begin{bmatrix} | & | & | \\ \lambda_1 v_1 & \lambda_2 v_2 & \lambda_3 v_3 \\ | & | & | \end{bmatrix}$$



$$Av_i = \lambda_i v_i$$

This means that v_i is an eigenvector of A with eigenvalue λ_i . Can v_i be zero? It cannot be zero as v_i is a column of an invertible matrix. This means that we are justified in calling v_i an eigenvector.

4. Diagonalizability: Definition-2

- P is an invertible matrix.
- The columns of P are eigenvectors of A .
- Therefore, there is a basis for $\mathbb{R}^n$ made up of eigenvectors of A .



Definition-2: A matrix A of shape $n \times n$ is diagonalizable if there is a basis for $\mathbb{R}^n$ made up of eigenvectors of A .

5. Eigen-decomposition

A diagonalizable matrix can be broken down into the product of three matrices which contain information about the eigenvalues and eigenvectors of the matrix:

$$A = PDP^{-1}$$

How do you diagonalize a matrix that is diagonalizable?

$$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$$

Step-1: Find the eigenvalues

$$\begin{aligned} |A - \lambda I| &= (1 - \lambda)(4 - \lambda) - 10 \\ &= \lambda^2 - 5\lambda - 6 \\ &= (\lambda - 6)(\lambda + 1) \end{aligned}$$

Therefore, the eigenvalues are $\lambda = 6, -1$. Always a good practice (convention) to list the eigenvalues in decreasing order.

Step-2: Find the eigenvectors

$$(A - 6I) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -5 & 2 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-5v_1 + 2v_2 = 0$$

$$v_1 = 2$$

$$v_2 = 5$$

Therefore, $(2, 5)$ is an eigenvector for A corresponding to $\lambda = 6$

Verify:

$$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix} \rightarrow 2(1, 5) + 5(2, 4) = (12, 30) = 6(2, 5)$$

$$E(6) = \text{span}\{(2, 5)\}$$

$$(A + I) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 2 \\ 5 & 5 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2v_1 + 2v_2 = 0$$

$(1, -1)$ is an eigenvector of A corresponding to $\lambda = -1$.

Verify:

$$(1, 5) - (2, 4) = (-1, 1) = -1(1, -1)$$

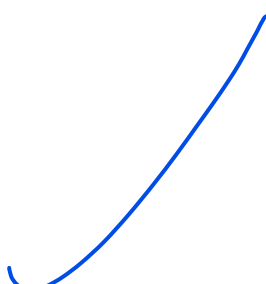
Step-3: Collect the eigenvalues in a diagonal matrix and the corresponding eigenvectors go into the columns of P

$$D = \begin{bmatrix} 6 & 0 \\ 0 & -1 \end{bmatrix}, \quad P = \begin{bmatrix} 2 & 1 \\ 5 & -1 \end{bmatrix}$$

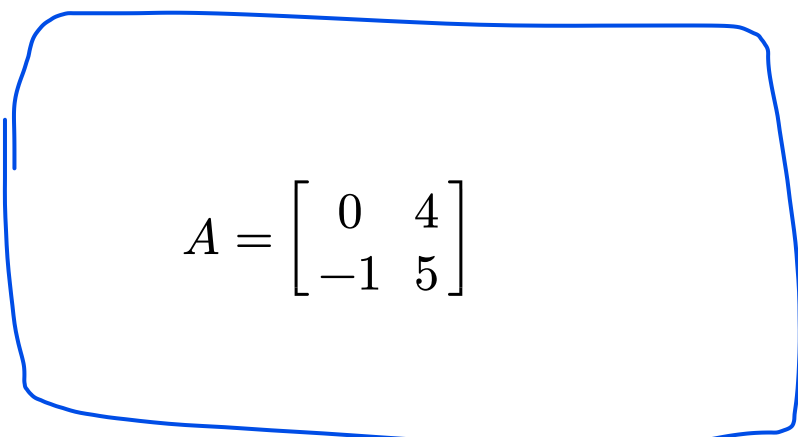
Step-4: Find P^{-1} and express $A = PDP^{-1}$

$$P^{-1} = \frac{1}{-7} \begin{bmatrix} -1 & -1 \\ -5 & 2 \end{bmatrix}$$

Verify:

$$\begin{aligned}PDP^{-1} &= \frac{1}{-7} \begin{bmatrix} 2 & 1 \\ 5 & -1 \end{bmatrix} \begin{bmatrix} 6 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ -5 & 2 \end{bmatrix} \\&= \frac{-1}{7} \begin{bmatrix} 12 & -1 \\ 30 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ -5 & 2 \end{bmatrix} \\&= \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix} \\&= A\end{aligned}$$


Homework


$$A = \begin{bmatrix} 0 & 4 \\ -1 & 5 \end{bmatrix}$$

6. Properties

In all these properties, A is some diagonalizable matrix:

$$A = PDP^{-1}$$

Matrix power

$$\begin{aligned}A^2 &= (PDP^{-1})(PDP^{-1}) \\&= PDDP^{-1} \\&= PD^2P^{-1}\end{aligned}$$

In general:

$$A^k = PD^kP^{-1}$$

Eigenvalues and eigenvectors

$$\begin{aligned} |A - \lambda I| &= |PDP^{-1} - \lambda I| \\ &= |PDP^{-1} - \lambda PIP^{-1}| \\ &= |P(DP^{-1} - \lambda IP^{-1})| \\ &= |P(D - \lambda I)P^{-1}| \\ &= |P| \cdot |D - \lambda I| \cdot |P^{-1}| \\ &= |D - \lambda I| \end{aligned}$$

A and D have the same characteristic polynomial and hence the same eigenvalues. Moreover, A has a full set of n eigenvalues.

Important: If an $n \times n$ matrix is diagonalizable, it has n eigenvalues, not necessarily distinct. When we say that there are n eigenvalues, we also include the multiplicity of eigenvalues. For example, for the identity matrix I of shape $n \times n$, there are n eigenvalues, but only one unique eigenvalue, namely 1, that is repeated n times.

True: If A is diagonalizable then A has n eigenvalues.

Invertibility

If A is invertible, then:

$$A^{-1} = PD^{-1}P^{-1}$$

Determinant

$$|A| = |PDP^{-1}| = |D|$$

Eigenvectors

Are the eigenvectors of A and D equal? Not necessarily.

- The eigenvectors of A can be read off by looking at the matrix P . The columns of P are the eigenvectors of A .
- The eigenvectors of D are the elements of the standard basis of $\mathbb{R}^n$.

7. Tests for Diagonalizability

How do we know if a given matrix is diagonalizable?

7.1. Negative Test

Consider the matrix A :

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$|A - \lambda I| = (1 - \lambda)^2 + 1$$

What are the eigenvalues? A doesn't have any real eigenvalues. It is alright to say that A doesn't have any eigenvalues while dealing with real vector spaces.

Remark: For weeks 3 and 4, we are only dealing with real vector spaces. Therefore, eigenvalues and eigenvectors, if they exist, have to be real.

Can A be diagonalizable? No. This is because if A had been diagonalizable, it would have had 2 eigenvalues.

Negative Test: If a matrix A of shape $n \times n$ doesn't have n eigenvalues (including repetitions) then A is not diagonalizable.

7.2. Positive Test

Consider a matrix A that has two distinct eigenvalues, say λ_1, λ_2 , where $\lambda_1 \neq \lambda_2$.

Theorem: Eigenvectors corresponding to distinct eigenvalues are linearly independent.

Let v_1, v_2 be eigenvectors corresponding to λ_1, λ_2 :

$$\alpha_1 v_1 + \alpha_2 v_2 = 0$$

$$(A - \lambda_1 I)(\alpha_1 v_1 + \alpha_2 v_2) = 0$$

$$\alpha_1(\lambda_1 - \lambda_1)v_1 + \alpha_2(\lambda_2 - \lambda_1)v_2 = 0$$

$$\alpha_2(\lambda_2 - \lambda_1)v_2 = 0$$

$\alpha_2 = 0$ because $\lambda_1 \neq \lambda_2$ and $v_2 \neq 0$ (v_2 is an eigenvector). If $\alpha_2 = 0$ then $\alpha_1 = 0$. Therefore, $\{v_1, v_2\}$ is linearly independent.

Positive Test: If a matrix A of shape $n \times n$ has n distinct eigenvalues, then A is diagonalizable.

Why?

- If there are n distinct eigenvalues, then the corresponding set of n eigenvectors are linearly independent.
- We now have a basis of eigenvectors for $\mathbb{R}^n$ and hence A is diagonalizable.

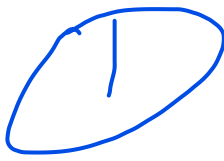
7.3. Inconclusive?

What if a matrix A of shape $n \times n$ has n eigenvalues where some of them repeat? Is such a matrix diagonalizable?

This is an inconclusive test. You can't comment on the diagonalizability of A .

8. Diagonalizable Matrices: Examples

Homework



$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$



$$A = \begin{bmatrix} 2 & 0 & -3 \\ 1 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix}$$

Step-1

$$\begin{aligned} |A - \lambda I| &= (-1 - \lambda)(2 - \lambda)(-1 - \lambda) \\ &= (2 - \lambda)(1 + \lambda)^2 \end{aligned}$$

Eigenvalues are $2, -1, -1$.

Step-2:

$$(A - 2I) \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} 0 & 0 & -3 \\ 1 & -3 & -1 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} \mathbf{1} & -3 & -1 \\ 0 & 0 & \mathbf{1} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

v_1 and v_3 are pivot variables and v_2 is a free variable.

$$v_1 = 3$$

$$v_2 = 1$$

$$v_3 = 0$$

$(3, 1, 0)$ is an eigenvector corresponding to $\lambda = 2$.

$$(A + I) \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} 3 & 0 & -3 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} \mathbf{1} & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = 0$$

$\lambda = -1$ will have two linearly independent eigenvectors.

A set of linearly independent eigenvectors for $\lambda = -1$:

$$\{(1, 0, 1), (1, 1, 1)\}$$

Therefore, a basis for $\mathbb{R}^3$ made up of eigenvectors of A :

$$\beta = \{(3, 1, 0), (1, 0, 1), (1, 1, 1)\}$$

Therefore A is diagonalizable.

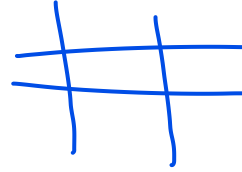
$$P = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & & \\ & -1 & \\ & & -1 \end{bmatrix}$$

Verify:

$$A = PDP^{-1}$$

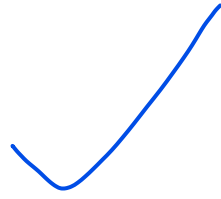
This is an eigen-decomposition of A .

9. Non-Diagonalizable Matrices



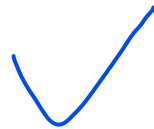
9.1. Example-1

$$A = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$$



Step-1: Find eigenvalues

$$\begin{aligned} |A - \lambda I| &= (3 - \lambda)(1 - \lambda) + 1 \\ &= \lambda^2 - 4\lambda + 4 \\ &= (\lambda - 2)^2 \end{aligned}$$



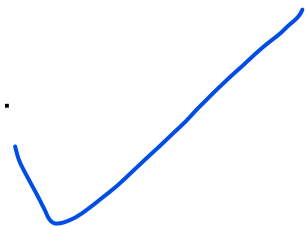
Step-2: Find the eigenvectors

$$(A - 2I) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



$(1, 1)$ is an eigenvector of A with eigenvalue $\lambda = 2$.



$$E(2) = \text{span}\{(1, 1)\}$$

- Can you find a basis of eigenvectors for $\mathbb{R}^2$?
- Can you find an invertible matrix whose columns are the eigenvectors of A ?

No

No

No, we can't.

$$P = \begin{bmatrix} 2 & 5 \\ 2 & 5 \end{bmatrix} \rightarrow \text{rank}(P) = 1$$

Conclusion is that $A = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$ is not diagonalizable.

9.2. Example-2

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

Step-1:

$$(2 - \lambda)^2 = 0 \\ \implies \lambda = 2$$

Step-2:

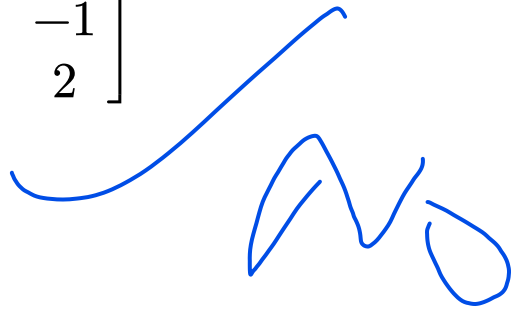
$$(A - 2I) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \\ \begin{bmatrix} 0 & \mathbf{1} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \implies v_1 = 1, v_2 = 0 \\ v_2 = 0$$

$(1, 0)$ is an eigenvector of A with eigenvalue $\lambda = 2$. You don't have a set of two independent eigenvectors for A , hence A is not diagonalizable.
There is no basis for the underlying linear transformation in which the matrix representation is diagonal.

9.3. Example-3

Homework

$$A = \begin{bmatrix} 5 & 1 & -1 \\ 1 & 3 & -1 \\ 2 & 0 & 2 \end{bmatrix}$$



10. Problems

10.1. Problem-1

Find A^k where:

$$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$$

Refer to section-5 for the eigen-decomposition of A :

$$D = \begin{bmatrix} 6 & 0 \\ 0 & -1 \end{bmatrix}, \quad P = \begin{bmatrix} 2 & 1 \\ 5 & -1 \end{bmatrix}, \quad P^{-1} = \frac{1}{-7} \begin{bmatrix} -1 & -1 \\ -5 & 2 \end{bmatrix}$$

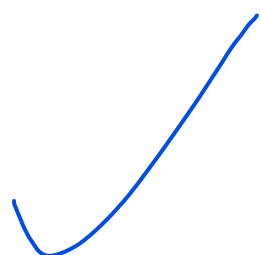
$$A = PDP^{-1}$$

$$A^k = PD^kP^{-1}$$

$$= \begin{bmatrix} 2 & 1 \\ 5 & -1 \end{bmatrix} \begin{bmatrix} 6^k & 0 \\ 0 & (-1)^k \end{bmatrix} \frac{1}{-7} \begin{bmatrix} -1 & -1 \\ -5 & 2 \end{bmatrix}$$

Relate this to Fibonacci sequence example:

$$\begin{bmatrix} F_{k+2} \\ F_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix}$$



Refer to this [document](#) for the details of the Fibonacci example discussed in the lectures.

11. Orthogonally Diagonalizable Matrices

Orthogonal matrices are square matrices Q that satisfy $Q^T Q = Q Q^T = I$. Therefore, $Q^{-1} = Q^T$. The columns of Q are orthonormal, they in fact form an orthonormal basis for $\mathbb{R}^n$.

Real Spectral Theorem: All symmetric matrices are diagonalizable. In fact, they are orthogonally diagonalizable. It means that if A is symmetric, then there is an orthogonal matrix Q and a diagonal matrix D such that:

$$A = Q D Q^T$$

This is often called the spectral decomposition.

Lemma: Eigenvectors corresponding to distinct eigenvalues of a symmetric matrix are orthogonal.

For week-5, not for quiz-1

Conjugate transpose: generalization of transpose to complex matrices:

$$A = \begin{bmatrix} 1+i & i \\ 2-i & 3i \end{bmatrix} \Rightarrow A^* = \begin{bmatrix} 1-i & 2+i \\ -i & -3i \end{bmatrix}$$

$$A^* = (\overline{A})^T$$

Hermitian matrix is one which is equal to its conjugate transpose matrix:

$$A^* = A$$

$$A = \begin{bmatrix} 1 & 1+i \\ 1-i & 2 \end{bmatrix}$$

$$A^* = \begin{bmatrix} 1 & 1+i \\ 1-i & 2 \end{bmatrix} = A$$

11.1. Problem-2

For what values of k is $A = \begin{bmatrix} 5 & 3 \\ k & -1 \end{bmatrix}$ diagonalizable over $\mathbb{R}$?

Answer: A is diagonalizable whenever $k > -3$.

Step-1: Find the characteristic polynomial

$$\begin{aligned} & (5 - \lambda)(-1 - \lambda) - 3k \\ &= \lambda^2 - 4\lambda - 5 - 3k \\ &= \lambda^2 - 4\lambda + (-5 - 3k) \end{aligned}$$

Step-2: Ensure real eigenvalues

$$\Delta = 16 + 4(5 + 3k)$$

For real eigenvalues, $\Delta \geq 0$:

$$\begin{aligned}\Delta &\geq 0 \\ 16 + 4(5 + 3k) &\geq 0 \\ 4 + 5 + 3k &\geq 0 \\ k &\geq -3\end{aligned}$$

If $k \geq -3$, A has two real eigenvalues. If the discriminant of a quadratic equation is greater than or equal to zero, then it has two real roots:

$$\lambda = \frac{4 \pm \sqrt{\Delta}}{2}$$

Step-3:

If $k > -3$, what can you say? A has two distinct eigenvalues, which means that A is diagonalizable.

Step-4:

Look at what happens when $k = -3$, repeated roots, or one eigenvalue repeated twice. Find this eigenvalue:

$$\lambda = 2, 2$$

$$(A - 2I)v = 0$$

$$\begin{bmatrix} 3 & 3 \\ -3 & -3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$(1, -1)$ is an eigenvector for A with eigenvalue for $\lambda = 2$. Since we don't have two linearly independent eigenvectors for A , A is not diagonalizable.